

# Examining LLM-Surprisal as an Indicator of Naturalness for Japanese Automated Essay Scoring

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## Abstract

This study investigates Large Language Model (LLM) surprisal as an indicator of global linguistic naturalness in L2 Japanese automatic essay scoring. Results demonstrate that surprisal effectively distinguishes learner proficiency levels. Combining surprisal with feature-based indices achieves the highest classification accuracy, validating surprisal as a valuable metric for automated writing evaluation.

## 1 Introduction

Automated Essay Scoring (AES) aims to provide scalable, efficient assessment of writing proficiency. Traditional AES typically relies on computational models trained on linguistic and statistical features, such as word counts, sentence length, and lexical diversity (Ramesh and Sanampudi, 2022). This feature-based paradigm remains dominant in Japanese as a second language (L2) assessment. For example, one of the most widely recognized Japanese AES systems, jWriter, primarily utilizes such feature-based linguistic indices (Lee and Hasebe, 2020).

While these feature-based approaches effectively capture surface-level textual properties, they fail to account for overall linguistic naturalness. Consequently, existing systems may struggle to reflect the holistic construct of writing proficiency evaluated by human raters.

Recent advances in natural language processing (NLP) and the rapid development of large language models (LLMs) have reshaped AES, frequently outperforming traditional feature-based models (Hirao et al., 2020; Li and Liu, 2024). Among various approaches, one of the promising directions is surprisal theory, rooted in psycholinguistics (Hale, 2001; Levy, 2008). Surprisal quantifies context predictability, with higher surprisal values indicating greater unexpectedness. From

a language processing perspective, lower average surprisal generally reflects more natural and fluent text (Cong, 2025), and correlates with writers' linguistic proficiency (Kharkwal and Muresan, 2014). For example, Cong (2025) reported that LLM-surprisal effectively captures naturalness, lexical diversity, and syntactic complexity in L2 English writing. Hu and Cong (2025, 2026) extended this to L2 Chinese, confirming its predictive power and cross-linguistic robustness across typologically distinct languages. These findings suggest that LLM-surprisal can serve as a theoretically grounded indicator of global linguistic naturalness in AES, effectively reflecting underlying lexical diversity and syntactic complexity.

Building on this literature, the present study investigates the role of LLM-surprisal in L2 Japanese AES, extending the application of surprisal-based metrics to a language that remains lower-resource compared to English. Japanese provides a compelling test case for surprisal's robustness due to its typologically distinct grammatical structure and complex orthography. By integrating LLM-surprisal with feature-based classic indices used in jWriter (Lee and Hasebe, 2020), we examine whether or not this combined approach enhances predictive performance. Replicating and extending Cong (2025) and Hu and Cong (2025, 2026), this study addresses the following research questions:

### (1) Proficiency sensitivity of LLM-surprisal.

Can LLM-surprisal reliably differentiate proficiency levels in L2 Japanese essays, including contrasts between L1 and L2 writing as well as across L2 subgroups (beginner, intermediate, advanced)?

### (2) Model variation in surprisal computation.

How do different language models vary in their calculation and distribution of surprisal scores, and to what extent do these differences affect their ability to distinguish proficiency

levels in L2 Japanese writing?

- (3) **Relationship between surprisal and classic indices.** To what extent are LLM-surprisal scores correlated with feature-based classic indices (e.g., lexical diversity, sentence length, morphosyntactic measures), and what unique variance does surprisal capture beyond these classic indices?
- (4) **Predictive contribution of surprisal in proficiency modeling.** How does the inclusion of surprisal measures in regression-based models influence the prediction of essay writers' proficiency levels compared to models relying solely on classic indices?

This study is significant in bridging the gap left by traditional, feature-based approaches to L2 Japanese AES. By integrating LLM-surprisal as an indicator of global naturalness, we investigate how this metric can enhance Japanese AES, offering insights for developing proficiency measurement tools that can support language learners.

## 2 Related work

### 2.1 Japanese AES

Research and development in Japanese AES has spanned several decades. One of the earliest systems was the Automated Japanese Essay Scoring System (JESS), developed by [Ishioka and Kameda \(2004\)](#) for Japanese native speakers. This program evaluates essays based on three primary features: rhetoric (lexical and vocabulary diversity and complexity), organization (use of rhetorical features and linguistic cues), and content (use of vocabulary related to the topic).

For L2 Japanese learners, [Lee and Hasebe \(2020\)](#) developed jWriter, which estimates essay proficiency levels using a linear regression model trained on linguistic features from the International Corpus of Japanese as a Second Language (I-JAS)<sup>1</sup> ([Sakoda, 2020](#)). Crucially, jWriter is widely accepted as a standard AES system for L2 Japanese, motivating us to integrate its classic indices as base features to explore the role of LLM-surprisal.

Following NLP advancements, researchers moved beyond classic indices. [Hirao et al. \(2020\)](#) developed GoodWriting Rater, a robust neural network AES system utilizing a BERT model. More recently, [Li and Liu \(2024\)](#) found that prompt-based LLM evaluation achieved higher

scoring accuracy than systems relying solely on classic indices like those in jWriter.

Despite these advances, research on LLM-based L2 Japanese AES remains limited. Moreover, many LLM scoring systems function as black boxes, making it difficult to interpret how specific linguistic properties influence predicted scores. For instance, [Obata et al. \(2023\)](#) found that adding ChatGPT scores with classic indices for L2 Japanese essays did not improve prediction accuracy. This suggests challenges regarding the practical reliability of the LLMs and highlights the robustness of traditional linguistic features. Consequently, there is a pressing need for transparent, theoretically grounded LLM-indices that clearly relate to L2 Japanese writing proficiency.

### 2.2 LLM-surprisal in AES

LLM-surprisal is the negative log probability of a word given its preceding context ([Cong, 2025](#); [Hale, 2001](#)). It strongly correlates with human language processing metrics like reading time; higher surprisal indicates greater processing difficulty, reflecting the heavier cognitive load required for unpredictable input ([Wilcox et al., 2023](#)).

In L2 AES, [Kharkwal and Muresan \(2014\)](#) first demonstrated that syntactic and lexical surprisal, computed via a syntactic parser ([Roark et al., 2009](#)), decreases as L2 English proficiency increases. More recently, [Tang and van Hell \(2024\)](#) confirmed that higher proficiency yields lower LLM-surprisal in L2 English writing. Furthermore, [Cong \(2025\)](#) showed that LLM-surprisal distinguishes between native (L1) and L2 English essays and differentiates L2 proficiency levels. Importantly, by analyzing its relationship with robust classic indices, [Cong \(2025\)](#) proposed LLM-surprisal as an indicator of L2 writing naturalness that numerically represents the interplay between lexical diversity and syntactic complexity.

Exploring contexts beyond English, [Hu and Cong \(2025, 2026\)](#) found LLM-surprisal effectively distinguishes L2 Chinese proficiency levels, demonstrating cross-linguistic applicability. The study also confirmed that LLM-surprisal may serve as a proxy for both linguistic complexity and the naturalness in L2 Chinese. Additionally, [Hu and Cong \(2025\)](#) reported that an LLM trained exclusively on traditional Chinese data outperformed multilingual or simplified Chinese models when evaluating essays written in traditional Chinese. This indicates that LLM training data coverage

<sup>1</sup><https://www2.ninjal.ac.jp/jll/lsaj/ihome2-en.html>

may significantly impact the utility of surprisals in AES.

### 3 Methods

#### 3.1 Data

The dataset for this study was sourced from the I-JAS, the L2 Japanese learner corpus that Lee and Hasebe (2020) utilized to train the jWriter model. This cross-sectional corpus comprises spoken and written data from 1,000 Japanese learners across 12 different native language backgrounds (Sakoda, 2020). Within the corpus, learner data is paired with their Simple Performance-Oriented Test (SPOT) score, which is a widely used standardized test that assesses learner’s overall Japanese proficiency (Kobayashi, 2016). While SPOT scores do not directly evaluate writing proficiency, they serve as a standard benchmark in Japanese AES research since the essays themselves lack individual scores (Lee and Hasebe, 2020; Li and Liu, 2024). Although other L2 Japanese writing corpora are available, none provide essay-level proficiency annotations that can serve as target scores for AES model development and evaluation. Consequently, SPOT scores are commonly used as a proxy for learner proficiency. The SPOT is scored out of a maximum of 90 points and categorizes learners into four estimated proficiency levels: Introductory (0–30), Beginner (31–55), Intermediate (56–80), and Advanced (81–90) (University of Tsukuba, 2013). Because the Introductory level indicates that the “learner has studied virtually no Japanese,” data from this level were excluded. The corpus also provides baseline data from native Japanese speakers, who were not assessed using the SPOT.

While the I-JAS includes multiple writing tasks (e.g., storytelling and email writing), this study specifically focused on an argumentative essay task requiring participants to write approximately 600 characters comparing fast food and home-cooked food. This yielded an initial pool of 683 essays.

Essays produced by learners scoring below the Beginner threshold, as well as extreme outliers in character count, were removed. Following this filtering process, 668 essays remained. To balance the dataset across proficiency levels and mitigate overrepresentation, a random subsample of 100 essays was drawn from the disproportionately large Intermediate group. This resulted in a final dataset of 329 essays ranging from 224 to 1,001 charac-

ters in length. The distribution and mean character count across proficiency levels are summarized in Table 1.

| Level        | N   | Mean   | Min | Max  | SD     |
|--------------|-----|--------|-----|------|--------|
| Beginner     | 93  | 578.57 | 224 | 949  | 128.06 |
| Intermediate | 100 | 609.66 | 345 | 990  | 112.84 |
| Advanced     | 88  | 619.77 | 257 | 1001 | 123.95 |
| Native       | 48  | 624.50 | 444 | 926  | 105.23 |

Table 1: Descriptive statistics of essay character length

Data preprocessing procedures included standardizing punctuation marks, eliminating irrelevant spaces and tabs, and removing annotator meta-data. Following this automated preprocessing, a native Japanese speaker who has systematic training in Japanese language conducted a manual screening of the dataset to verify the formatting of the essays.

#### 3.2 Classic indices

This study utilized a set of classic indices identified as the most significant predictors in the jWriter regression model (Lee and Hasebe, 2020), which are also widely studied in AES (Kyle and Crossley, 2015, 2018). These features include average sentence length (mean number of tokens per sentence), type-token ratio (the ratio of unique word types to the total number of word tokens, serving as a measure of lexical diversity), number of verbs, number of lower-intermediate words, number of upper-intermediate words, total number of characters, number of Japanese-originated words (*wago*), and number of Chinese-originated words (*kango*). We programmatically extract these classic indices, leveraging the MeCab Version 0.996 morphological analyzer (Kudo, 2005) in conjunction with the IPADIC Version 1.0.102 dictionary. Vocabulary difficulty levels, as well as the identification of *wago* and *kango*, were determined by cross-referencing a Japanese educational vocabulary database (Sunakawa et al., 2012).

#### 3.3 LLM-Surprisal indices

LLM-surprisal is formally defined as the negative log probability of a word or a sequence of words given its preceding context, as computed by language models (Cong, 2025; Hale, 2001). It reflects the predictability of a sequence: less predictable sequences receive higher surprisal values and are often associated with lower acceptability.

Consider the following sentences:

- (1) 私は本を 読みました。(を's surprisal:1.93)  
 watashi-wa hon-o yomi-mashi-ta  
 I-TOP book-ACC read-POL-PAST  
 'I read a book.'
- (2) \* 私は本が 読みました。(が's surprisal:4.43)  
 watashi-wa hon-ga yomi-mashi-ta  
 I-TOP book-NOM read-POL-PAST  
 'I read a book.'

The two Japanese sentences form a minimal contrast, differing only in the case particle attached to 本. In (1), the accusative marker を appropriately marks the object of 読みました, whereas (2) uses が, which makes the sentence ungrammatical. For each sentence, we computed the surprisal value for each token conditioned on its preceding context using the LLaMA-3.1 model (Meta, 2024). The model assigns lower surprisal to を (1.93) and higher surprisal to が (4.43), indicating that が is less predictable in this syntactic context. This contrast illustrates surprisal's sensitivity to Japanese case-marking violations.

In the current study, since surprisal estimation is sensitive to variation in text length (Rezaii et al., 2023), essays longer than 600 characters were truncated only for the surprisal analysis<sup>2</sup>. We selected this threshold because it approximates the mean character count across proficiency levels (see Table 1). For each essay, token-wise surprisal values were computed and averaged across tokens to obtain a mean surprisal score (see Appendix E for examples). Since our curated dataset is controlled in topic, content, and writing style (c.f., argumentative essay task only), we maintain that isolated from confounding factors in open-ended, unstructured narratives, lower surprisal values derived from our dataset are interpreted as reflecting more natural and target-like language use, which may be associated with higher writing proficiency. Accordingly, we hypothesize that surprisal values will decrease as learner proficiency increases.

### 3.4 LLMs selection

We computed surprisal using the `minicons` toolkit (Misra, 2022), which provides a standardized interface for extracting token-level probabilities from LLMs in Python. We examined three autoregressive large language model families: *Llama-3.1-8B models* (**LLaMA Base/Instruct**)

<sup>2</sup>We also conducted additional surprisal analysis using the full, untruncated essays and obtained a highly similar pattern of results, suggesting that the findings were not affected by the truncation procedure.

(Grattafiori et al., 2024), Japanese-enhanced *Llama-3.1-Swallow-8B-v0.5 models* (**TokyoLM Base/Instruct**) (TokyoTech LLM Team, 2025) and Japanese-specific *llm-jp-3-3.7b models* (**JP-LLM Base/Instruct**) (Research and Development Center for Large Language Models, National Institute of Informatics, 2025). The TokyoLM models were further pretrained on large-scale Japanese corpora, whereas the JP-LLM models were trained primarily on Japanese data. Detailed model information is provided in Appendix A. These models vary along two dimensions: training data composition (multilingual, Japanese-enhanced, and Japanese-specific) and training objective (base pretraining vs. instruction-tuned), allowing us to examine how these factors influence the models' behavior and surprisal estimates. All models are publicly available on [HuggingFace](https://huggingface.co).

### 3.5 Data Analysis

All analyses were conducted in R (version 4.5.1, R Core Team, 2025)<sup>3</sup>. We first examined whether LLM-surprisal differentiated native and learner writing to establish a baseline. We then assessed whether surprisal could further distinguish writing across L2 proficiency levels. For each of the six language models, essay-level surprisal scores were compared between L1 and L2 writers using nonparametric Wilcoxon tests. Differences across L2 proficiency groups (beginner, intermediate, advanced) were then evaluated using Kruskal-Wallis tests, followed by pairwise Wilcoxon rank-sum tests with Bonferroni correction when the overall effects were significant. Effect sizes were further compared across models to assess differences in proficiency sensitivity.

Next, we conducted correlation analyses to examine the relationship between LLM-surprisal and (i) overall L2 proficiency (SPOT scores), as well as (ii) classic linguistic indices (see Section 3.2). These analyses assessed the extent to which surprisal was associated with established feature-based measures of Japanese L2 writing.

Finally, we evaluated the predictive contributions of surprisal using multinomial logistic regression models for L2 proficiency classification (beginner, intermediate, and advanced). Three feature configurations were compared: (i) surprisal indices only, (ii) classic indices only, and (iii) a combined feature set including both. The features

<sup>3</sup>The code is publicly available at <https://github.com/JingyingHu/Japanese-essays-surprisal>

included in each configuration and the corresponding model specifications are provided in Appendix B. This controlled ablation study (with different feature combinations) allowed us to quantify the change of classification performance brought by different feature sets and pinpoint the predictive contribution of surprisal. All predictors were z-standardized prior to modeling. Model performance was evaluated using repeated stratified 5-fold cross-validation, repeated 10 times. For each fold, models were trained on the training partitions and evaluated on held-out testing partitions to reduce overfitting and assess robustness. Predictive performance was assessed using classification accuracy, precision, recall, and F1 scores.

## 4 Results

### 4.1 Proficiency sensitivity of LLM-surprisal

**L1–L2 contrasts** Across all six language models, L2 essays received higher surprisal scores than L1 essays did (see Figure 1). Wilcoxon rank-sum tests confirmed that these differences were significant for all models (all  $p$ s < .01; see Appendix C). This pattern indicates that learner writing was generally less “predictable” under the language models than native writing. The effect sizes also varied by model: the Japanese-specific JP-LLM models showed the strongest L1–L2 differentiation ( $r$ s = .47–.48), followed by the Japanese-enhanced TokyoLM models ( $r$ s = .35–.37), whereas the multilingual LLaMA models showed smaller effects ( $r$ s = .17–.19).

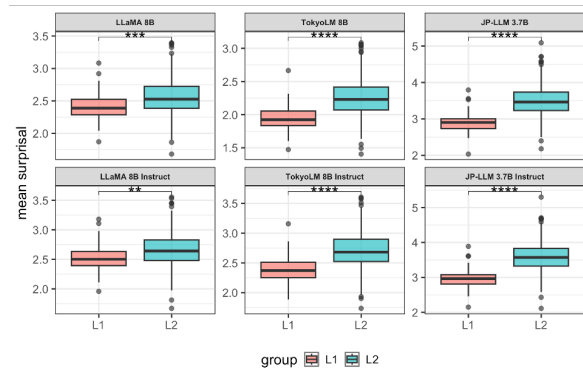


Figure 1: L1–L2 comparisons of LLM-surprisal scores across six language models. Significance levels: \*\*  $p$  < .01, \*\*\*  $p$  < .001, \*\*\*\*  $p$  < .0001.

**Within-L2 proficiency contrasts** Figure 2 shows surprisal differences across L2 proficiency levels. As predicted, surprisal scores generally

decreased as L2 proficiency increased across all models. Kruskal–Wallis tests showed significant proficiency effects for all models (all  $p$ s < .001; see Appendix D). Pairwise comparisons further showed that the Japanese-specific JP-LLM models most consistently differentiated all three proficiency levels, whereas the multilingual LLaMA models showed weaker contrasts, particularly between the beginner and intermediate groups. Effect sizes were again largest for the JP-LLM models ( $\eta^2$ s = .32–.34), followed by the TokyoLM models ( $\eta^2$ s = .14–.20), and smallest for the LLaMA models ( $\eta^2$ s = .07).

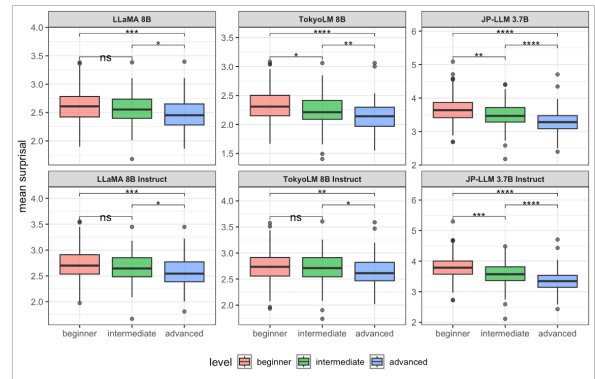


Figure 2: Comparison of mean LLM-surprisal scores across three L2 proficiency levels. Significance levels: ns (not significant), \*  $p$  < .05, \*\*  $p$  < .01, \*\*\*  $p$  < .001, \*\*\*\*  $p$  < .0001.

### 4.2 Relationship between surprisal and classic indices

Figure 3 presents Spearman correlations between LLM-surprisal scores, SPOT scores, and classic indices in L2 Japanese essays. Total character count was excluded because surprisal was computed from truncated essays. Across all models, surprisal scores were negatively correlated with SPOT scores, indicating that higher-proficiency learners generally produced lower-surprisal writing. The strongest associations were observed for the Japanese-specific JP-LLM models ( $r$ s = −0.43 and −0.39), whereas the multilingual LLaMA models showed comparatively weaker correlations (both  $r$ s = −0.23).

### 4.3 Predictive contribution of surprisal in proficiency modeling

Table 2 summarizes the cross-validation performance of the multinomial logistic regression models across the three feature configurations on held-out testing data. Overall, the combined model

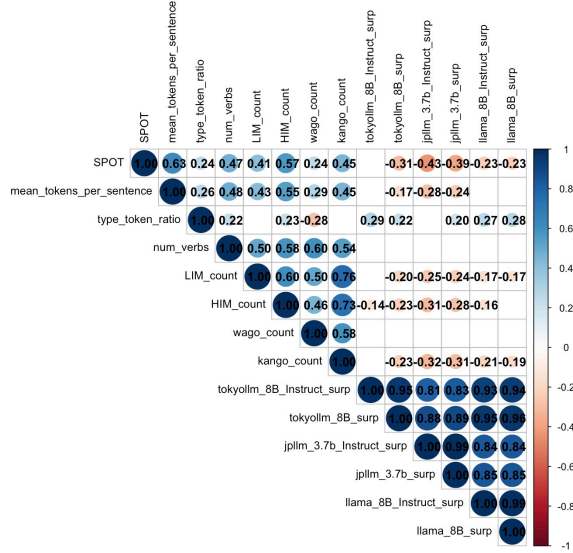


Figure 3: Heatmap of correlation coefficients between LLM-surprisal scores, classic linguistic indices, and SPOT scores. Insignificant cells are left blank.

achieved the strongest performance across all evaluation metrics, including accuracy, precision, recall, and F1. Importantly, the surprisal-only model consistently outperformed the classic-indices-only model, suggesting that LLM-derived surprisal captures proficiency-related information beyond traditional feature-based approaches. These findings highlight the potential of LLM-derived surprisal as a complementary indicator for improving automated L2 proficiency assessment.

| Models    | Classic only  | Surprisal only | Combined      |
|-----------|---------------|----------------|---------------|
| Accuracy  | 0.591 (0.057) | 0.603 (0.066)  | 0.617 (0.055) |
| Precision | 0.599 (0.059) | 0.611 (0.065)  | 0.626 (0.056) |
| Recall    | 0.595 (0.057) | 0.606 (0.066)  | 0.621 (0.055) |
| F1        | 0.592 (0.057) | 0.604 (0.066)  | 0.618 (0.055) |

Table 2: Cross-validation performance of multinomial logistic regression models across three feature configurations. Values are reported as mean (SD). Model specifications are provided in Appendix B.

## 5 Discussion and Conclusion

This study investigated the potential of LLM-surprisal as an indicator of global linguistic naturalness in L2 Japanese essays. The findings demonstrate that surprisal effectively distinguishes between L1 and L2 writing, and it differentiates the internal proficiency levels of L2 learners with 60.3% accuracy, which resonates with and expands Cong (2025) and Hu and Cong (2025, 2026) to Japanese.

Regarding the variations among language models, the results highlight the importance of utilizing models trained specifically on the target language, directly mirroring the findings of Hu and Cong (2025, 2026) in the L2 Chinese context. The Japanese-specific model, JP-LLM, exhibited superior sensitivity compared to the multilingual LLaMA baseline. Besides language-specific coverage in the model pretraining stage, another possible explanation for this divergence lies in the differences in tokenizer granularity across models. Because we used model-native tokenizers, part of the observed advantage of Japanese-specific models may reflect tokenization compatibility with the Japanese writing system rather than purely superior sensitivity to learner proficiency. This suggests that a language model must possess robust and specific representations of Japanese phrasing to accurately assess the naturalness of L2 structures.

When comparing the base model with its instruction-tuned counterpart across the three model families, the results were mixed. The TokyoLM Base significantly outperformed its instruction-tuned version, which may reflect the tendency of base models to capture raw statistical distributions and exhibit high sensitivity to structural typicality. Conversely, within a model family, instruction tuning optimizes for task completion, which can shift internal probabilities and dilute sensitivity to learner grammar. However, the JP-LLM Instruct slightly exceeded its base equivalent. This outcome suggests that for models heavily optimized for Japanese, task tuning does not sacrifice sensitivity to linguistic naturalness.

The correlation analysis provided valuable insight into how surprisal interacts with classic linguistic indices. LLM-surprisal showed negative correlations with the SPOT score, average sentence length, *kango* count, and both upper and lower intermediate vocabulary. This indicates that as learners become more proficient, they use more advanced words in longer sentences; they use those advanced words in idiomatic, native-like collocations, thereby reducing the overall surprisal. This is in line with the argument by Cong (2025) that surprisal can serve as a proxy for naturalness, representing the interplay between the lexical diversity and syntactic complexity of the essay. Interestingly, the type-token ratio showed a weak positive correlation with surprisal. This suggests that a higher variety of vocabulary does not guarantee

contextual mastery, as learners may introduce a wide array of words without fully mastering how they fit together naturally in context.

Finally, the regression modeling confirmed that the classic indices utilized in systems like jWriter (Lee and Hasebe, 2020) already provide a highly robust foundation for proficiency classification. The surprisal-only model nevertheless slightly outperformed these traditional indices, demonstrating that LLM-surprisal alone possesses strong predictive power for writer proficiency. Building upon the robust foundation of the classic indices, the inclusion of surprisal yields the highest classification performance, increasing overall accuracy from 59.1% to 61.7%. Although the numerical gain is modest, it is notable given that the models predict overall learner proficiency from a single text sample. This improvement indicates that LLM-surprisal provides a distinct measure of global linguistic naturalness that traditional indices may not fully account for. Together, they capture unique variances and form a comprehensive evaluation framework.

In conclusion, this study establishes LLM-surprisal as a valuable and theoretically grounded metric for L2 Japanese AES. By capturing elements of writing naturalness and contextual predictability often overlooked by traditional systems, surprisal provides additional predictive power for automated writing evaluations. Integrating language-specific LLM-surprisal with classic linguistic indices offers a promising path forward for developing more holistic and effective educational measurement tool for Japanese language learners.

## Limitations

Several limitations should be noted regarding the current study. First, the analysis relied on the SPOT score, which evaluates the general Japanese proficiency of the writer rather than the holistic quality of the specific essay evaluated. Future research should incorporate human rater scores for the individual essays to better align the surprisal metrics with actual writing performance. Second, the current methodology calculates surprisal exclusively at the token-level. While this effectively captures local predictability, it may overlook broader structural phenomena. Subsequent studies should explore sentence-level and discourse-level surprisal indices to provide a more comprehensive evaluation of narrative flow and cohesion. Third,

the analysis was restricted to a single argumentative essay topic comparing fast food and home-cooked meals. Examining a broader range of writing tasks and genres will help confirm the stability of surprisal metrics across different contexts. Fourth, the dataset was relatively small following the downsampling procedure utilized to balance the proficiency classes. Larger-scale studies are needed to further validate the regression weights. Finally, we only focused on autoregressive models; investigating masked language models could provide additional perspectives on bidirectional contextual predictability in L2 Japanese essays. Despite the limitations, our work contributes to the research that language-specific LLMs provide a more effective and cross-linguistically robust approach to AES for Japanese learners.

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## A Language model specifications

| Model                | Type              | Training |
|----------------------|-------------------|----------|
| LLaMA 8B             | Multilingual      | Base     |
| LLaMA 8B Instruct    | Multilingual      | Instruct |
| TokyoLM 8B           | Japanese-enhanced | Base     |
| TokyoLM 8B Instruct  | Japanese-enhanced | Instruct |
| JP-LLM 3.7B          | Japanese-specific | Base     |
| JP-LLM 3.7B Instruct | Japanese-specific | Instruct |

Note. Language models used for surprisal estimation.

## B Feature Sets and Model Specifications

| Component                  | Variables                                               |
|----------------------------|---------------------------------------------------------|
| Classic linguistic indices | Mean tokens per sentence                                |
|                            | Type-token ratio                                        |
|                            | Number of verbs                                         |
|                            | Total character count                                   |
|                            | LIM count                                               |
|                            | HIM count                                               |
|                            | Wago count                                              |
| Surprisal indices          | Kango count                                             |
|                            | TokyoLLM-8B-Instruct surprisal                          |
|                            | JPLLM-3.7B-Instruct surprisal                           |
|                            | JPLLM-3.7B surprisal                                    |
|                            | TokyoLLM-8B surprisal                                   |
|                            | Llama-8B surprisal                                      |
| Outcome                    | Llama-8B-Instruct surprisal                             |
|                            | L2 proficiency level (beginner, intermediate, advanced) |

The three model specifications were as follows:

- **Classic-only model:** Level ~ Classic linguistic indices
- **Surprisal-only model:** Level ~ Surprisal indices
- **Combined model:** Level ~ Classic + Surprisal indices

## C Wilcoxon Rank-Sum test results for L1 vs. L2 group surprisal scores across LLMs

| Model                | $W$    | $p$    | $r$ |
|----------------------|--------|--------|-----|
| LLaMA 8B             | 4689.5 | < .001 | .19 |
| LLaMA 8B Instruct    | 4838.5 | .002   | .17 |
| TokyoLM 8B           | 2618.5 | < .001 | .37 |
| TokyoLM 8B Instruct  | 2834.0 | < .001 | .35 |
| JP-LLM 3.7B          | 1526.0 | < .001 | .47 |
| JP-LLM 3.7B Instruct | 1489.0 | < .001 | .48 |

Note.  $W$  = Wilcoxon rank-sum test statistic;  $r$  = effect size.

## D Kruskal–Wallis test results for surprisal scores across L2 proficiency levels

| Model                | $\chi^2(2)$ | $p$    | $\eta^2$ |
|----------------------|-------------|--------|----------|
| LLaMA 8B             | 24.97       | < .001 | .07      |
| LLaMA 8B Instruct    | 24.11       | < .001 | .07      |
| TokyoLM 8B           | 66.82       | < .001 | .20      |
| TokyoLM 8B Instruct  | 49.01       | < .001 | .14      |
| JP-LLM 3.7B          | 105.65      | < .001 | .32      |
| JP-LLM 3.7B Instruct | 114.67      | < .001 | .34      |

Note.  $\chi^2$  = Kruskal–Wallis test statistic;  $\eta^2$  = effect size.

## E Text excerpts and average surprisals by proficiency level

### Native (SPOT: N/A)

TokyoLM 8B (base = 1.842; instruct = 2.286), JP-LLM 3.7B (base = 2.556; instruct = 2.58), LLaMA 8B (base = 2.197; instruct = 2.286)  
 ファースト・フードと家庭料理、このふたつは別のカテゴリーに分類されるように見えるが、忙しい現代生活を生きる私たちは、少なくとも月に何回か、または年に何回か、人によっては週に何回もファースト・フードに頼っており、私たちの食生活と健康を考える上では「食生活」という大きなカテゴリーの中に共存するものとして捉えることができると思う。

### Advanced (SPOT: 83)

TokyoLM 8B (base = 1.856; instruct = 2.467), JP-LLM 3.7B (base = 3.05; instruct = 3.096), LLaMA 8B (base = 2.167; instruct = 2.272)  
 社会経済の発展と生活水準の向上につれて (collocation)、私たちの食生活も豊かになりました。その一つの特徴は外食産業の発達です。昔はずっと家で手料理を食べる (aspect/tense) のに対して、今はよくファーストフードを食べるようになってきました。手料理とファーストフードはそれぞれに優劣があると私はそう (redundancy) 思います。

### Intermediate (SPOT: 69)

TokyoLM 8B (base = 2.176; instruct = 2.608), JP-LLM 3.7B (base = 3.63; instruct = 3.751), LLaMA 8B (base = 2.444; instruct = 2.553)  
 「食事何 (particle) をしますか。ファーストフードですか。家で作った食べ物ですか。それはなぜですか。」この質問考えた (particle) ことがありますか。考えたことがないなら今更 (lexical choice) 考えるほうがいいと思います。何故ならば栄養は食事からとります (sentence structure)。それで栄養は健康のため大事なんです。

### Beginner (SPOT: 50)

TokyoLM 8B (base = 2.596; instruct = 3.022), JP-LLM 3.7B (base = 4.712; instruct = 4.684), LLaMA 8B (base = 3.01; instruct = 3.159)  
 ファーストフードと、手作り料理とどちら (particle) いちばん (lexical choice) すきですか (writing style)。今日はマクドナルドが世界中で知られている。世界中で、人々にビッグマク (spelling) を (particle) 食べられる (aspect)。でも健康よう (conjugation) に、毎日その食べ物食べない (particle) で！手作り料理はファーストフードより栄養価が高いだ (conjugation)。

Note. Underlined words indicate unnatural or incorrect usage, and the error category is provided in parentheses. Readers may refer to the source I-JAS dataset for full-length text.